Additional Material: Seeing the Process Through the Worker's Eyes: Event Logs from Egocentric Factory Videos

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This document accompanies the main article. It contains Appendices A to F. References to numbered sections, figures, tables, and equations without an appendix prefix refer to the main article. The appendices preserve the existing evidence, judgments, and prompt templates.

Appendix A. Journal Extension and Evidence Record

Table A1 compares the supplied short manuscript with this revision. It documents the extension of that manuscript; it does not assert a publication status for a workshop version. No new extraction experiment or audit has been added. The complete task-to-deliverable and claim-to-evidence registers are in `paper/revision/`.

Table A1. Workshop-to-journal revision map relative to the supplied short manuscript.

Existing material	Journal addition	Type of contribution
Five-stage overview	Formal event model, implementation algorithm, stage failures, guarantee table	Method specification
Brief time description	Synthetic-time equation, case comparison, timestamp and inclusion manifests	Measurement analysis and reporting correction
Category percentages	Activity/context/category distinction and exact transport example	Methodological finding from existing artifacts
Coverage caveats	Missing-time proposition with explicit assumptions	New analytical reasoning
Short foundry illustration	Transcript timeline, reconstructed event, provenance chain	Evidence-traceability example
Condensed audit	Rubric, denominators, ambiguity examples, all judgments	Fuller reporting of existing assessment
Numerical results	Portable recalculation, manifests, corrected terminology, six cases	Reproducibility and interpretation
Brief limitations	Observability taxonomy, uncertainty path, operational hypotheses, responsible use	Methodological synthesis

Appendix B. Reproducibility and Configuration

The offline command below reads the five frozen artifact directories and the original category map. It writes derived CSVs, a JSON summary, and typeset appendices to a separate output directory. It requires Python 3 and no model credentials or third-party packages.

```
python3 paper/reproduce.py
# Optional explicit locations:
python3 paper/reproduce.py --root /path/to/repository \
  --output-dir /path/to/results
```

The output includes `artifact_manifest.csv` (path, size, SHA-256), `inclusion_manifest.csv` (all 322 decisions), `transcript_intervals.csv`, `structural_metrics.csv`, `factory_metrics.csv`, `worker_metrics.csv`, `worker_comparisons.csv`, `factory003_transport.csv`, and `audit_judgments.csv`. The summary records script and prompt hashes. Merged-log consistency is checked separately by reconstructing S5 output in memory and comparing it to every released merged file. Source artifacts remain unchanged.

The original pandas-based assessment and value-analysis scripts accept portable output paths. The original activity-level discovery script retains its miner, noise threshold 0.2, and video case rule. Running it requires the dependencies documented in `paper/requirements-assessment.txt`. The environment used for this revision is recorded in `paper/revision/assessment_environment.txt`; it is not claimed to be the original environment. Table A2 maps paper results to their calculations.

Table A2. Reproduction map. Paths for the original scripts are relative to `paper/docs/assessment_materials/`.

Paper result	Calculation and frozen input	Saved evidence
Tables 7, 8, 11	`reproduce.py`; source texts and per-video logs; `assess.py` for original diagnostics	Artifact, inclusion, interval, and structural manifests
Tables 13, 14 and factory cases	`reproduce.py` and `value_analysis.py`; original process map and retained rows	Factory metrics, worker metrics, comparisons, composition totals
Figure 6	`reproduce.py`; exact duration totals and original map	Generated TikZ bars and category percentages
Figure 2 and Section 8.1	Activity-label filter over factory 003 CSVs	101 rows in `factory003_transport.csv`
Table 12, Appendix E	Mechanical transcription of `audit_verdicts.md`	84 judgments in CSV and LaTeX
Section 7.4.2	Original `pm_utility.py`, all per-video rows	`activity_discovery.json` and execution output
Equation 4	Algebra under explicitly idealized assumptions	Proof in Section 8.3; no empirical bound

Regeneration is a separate workflow using scripts 01 to 05 in order. Paths and model identifiers are command-line options in the extraction scripts. S1 requires source video archives and a Gemini credential; S2 and S3 use saved text and Gemini; S4 uses saved text, vocabularies, and an OpenAI credential; S5 is local. Their default model strings are `gemini-3.1-pro-preview` and `gpt-5.4`. S4 configures high reasoning effort and a strict JSON response schema; S1 to S3 do not explicitly configure temperature, a seed, or generation-token limits. Transient HTTP status codes 408, 429, 500, 502, 503, and 504 trigger retries; the scripts configure up to five attempts with increasing delay. S1 also has an outer per-video retry loop. These are code settings, not verified execution totals.

The archive does not provide per-request model responses, usage records, exact execution dates, token counts, costs, runtimes, historical dependency versions, or a resolved source revision. Existing outputs may be reused without a content-validity check. Consequently, changing a prompt while retaining existing outputs does not prove that those outputs were generated with the changed prompt. Frozen artifacts, hashes, and the remaining historical queries are recorded so that this limitation is explicit.

Appendix C. Activity-Level Discovery Recalculation

Table A3. Recalculation with the original activity-level discovery settings. Fitness is average trace fitness from token replay; precision is token-based precision.

Factory	Cases	Fitness	Precision
001	97	0.9801	0.0691
002	54	0.9910	0.0603
003	72	0.9874	0.0906
004	51	0.9823	0.0666
005	22	0.9544	0.2679
006	26	0.9720	0.0832

PM4Py version for this recalculation: 2.7.23.11. The original settings are retained. The saved result records the current PM4Py version. No archived process-level computation or result was found, so no process-level fitness or precision value is reported. Discovery-model images are omitted because the original models were not preserved. The repeated calculation establishes the shown model-log metrics under the recorded current environment, not a recovery of the original runtime.

Appendix D. Complete Operational-Category Map

This table reproduces the original map with factory scope made explicit. VA denotes production/completion work under the authors' convention, MH material handling, TR transport, QI inspection, CM cleaning/maintenance, RW rework, and DOC documentation. The sole unmapped label is unused in the archived events.

Factory	Process label	Category
001	Metal stamping	VA
001	Metal forming	VA
001	Manual mechanical assembly	VA
001	Semi-automated press assembly	VA
001	Structural frame assembly	VA
001	Component bagging	VA
001	Component kitting	VA
001	Material replenishment	MH
001	Finished goods transport	TR
001	Scrap disposal	MH
001	Workstation cleaning	CM
001	Production tracking	DOC
001	Machine maintenance	CM
002	Fabric untangling	MH
002	Fabric edge alignment	VA
002	Pattern matching	VA
002	Fabric folding	VA
002	Fabric pairing	VA
002	Fabric pinning	VA
002	Sub-assembly marking	VA
002	Fabric bundling	VA
002	Garment ironing	VA
002	Garment quality inspection	QI
002	Thread snipping	VA
002	Garment tagging	VA
002	Hanger assembly	VA
002	Garment bagging	VA
002	Cardboard sleeve packaging	VA
002	Cardboard wrapping	VA
002	Package labeling	VA
002	Master bagging	VA
002	Box assembly	VA
002	Box packing	VA
002	Box sealing	VA
002	Cart loading	TR
002	Material transport	TR
002	Floor auditing	DOC
003	Metal casting separation	VA
003	Wax pattern assembly	VA

Factory	Process label	Category
003	Surface finishing and polishing	VA
003	Defect patching	RW
003	Component sub-assembly	VA
003	Injection molding	VA
003	Mold preparation and cleaning	CM
003	Machine pressing	VA
003	Spot painting	VA
003	Thermal welding	VA
003	Material sorting and sifting	MH
003	Inventory verification	DOC
004	Fabric spreading	VA
004	Bulk fabric cutting	VA
004	Fabric measuring	VA
004	Overlock seaming	VA
004	Continuous fabric chaining	VA
004	Edge binding attachment	VA
004	Piping attachment	VA
004	Curvilinear seam joining	VA
004	Tape feeding	VA
004	Heat press labeling	VA
004	Garment hemming	VA
004	Thread trimming	VA
004	Quality inspection	QI
004	Garment folding	VA
004	Garment ironing	VA
004	Garment bagging	VA
004	Bulk box assembly	VA
004	Pick-and-pack sorting	MH
004	Production data logging	DOC
004	Scrap material management	MH
004	Machine maintenance	CM
004	Material transport	TR
005	motor assembly finishing	unmapped
005	motor labeling and nameplate attachment	VA
005	motor packaging and strapping	VA
005	automated stator winding	VA
005	semi-automatic stator wire insertion	VA
005	stator coil binding	VA
005	stator lead wire finishing and soldering	VA
005	manual lathe machining	VA
005	CNC machine tending	VA
005	drill press machining	VA
005	milling machine operation	VA
005	component deburring and polishing	VA
005	dimensional quality inspection	QI
005	manual hardware pre-assembly	VA
005	batch pneumatic fastening	VA
005	material transport and logistics	TR
005	part sorting and staging	MH
005	finished goods packaging	VA
006	Sand mold creation	VA
006	Pattern cleaning and coating	VA

Factory	Process label	Category
006	Core and insert placement	VA
006	Mold assembly and closing	VA
006	Mold track advancement	MH
006	Machine cleaning and maintenance	CM
006	Manual production logging	DOC
006	Wooden component scraping	VA
006	Wooden component assembly	VA
006	Machine sanding	VA
006	Powder sifting	VA
006	Chemical dough mixing	VA
006	Dough kneading	VA
006	Machine extrusion	VA
006	Punch press operation	VA
006	Bulk material transport	TR
006	Component staging and arrangement	MH
006	Floor-based part sorting	MH
006	Part mechanical modification	VA
006	Compression molding	VA
006	Raw material mixing	VA
006	Post-processing edge finishing	VA
006	Metal casting separation	VA
006	Workspace clearing	CM

Appendix E. Complete Archived Audit Judgments

The following tables preserve all 84 verdicts and source identifiers. The shorthand `001/w003/v00009`, for example, identifies factory 001, worker 003, recording 00009. Event verdicts are C (fully supported), P (partially supported), and X (contradicted). Entry verdicts are COV (overlap), EXC (uncovered and judged non-operational), and MISS (uncovered and judged operational). The archive's unsupported truncation explanation is flagged in entry 10. Punctuation is normalized for typesetting; judgments are unchanged.

Appendix E.1. Archived event judgments

No.	Source	Interval; verdict	Label and archived note
1	001/w003/v00009	00:01:38-00:02:12; C	mate components over central jig; assembly over metal block
2	001/w002/v00001	00:08:41-00:08:48; C	drop sealed bags into bulk container
3	001/w006/v00000	00:07:50-00:08:02; C	receive part from adjacent coworker
4	001/w005/v00006	00:17:09-00:17:15; P	actuate machine press; interval covers full load-press-remove cycle
5	001/w005/v00004	00:14:00-00:14:55; P	place main component into larger bag; transcript: "repetitive assembly and packaging continues"
6	001/w004/v00000	00:06:34-00:06:43; P	drive fastener into metal parts; transcript: "repetitive assembly continues"
7	001/w003/v00006	00:04:15-00:06:22; C	insert small pin into base components; split of long composite entry
8	002/w009/v00000	00:19:07-00:20:00; C	align printed fabric patterns
9	002/w006/v00002	00:05:39-00:06:11; C	enclose garment in pre-printed cardboard folder
10	002/w002/v00007	00:10:06-00:10:18; C	attach black plastic hanger hook to bag
11	002/w007/v00002	00:08:16-00:08:31; C	wrap white cardboard band around folded shirt
12	002/w005/v00007	00:06:09-00:06:24; C	fold t-shirt over removable white template; split of repeated-cycle entry
13	002/w002/v00001	00:12:22-00:12:26; P	drop packaged garments into bulk box; transcript: "appearing to drop an item"
14	002/w002/v00001	00:11:41-00:11:43; C	drop packaged garments into bulk box
15	003/w003/v00000	00:10:36-00:13:22; C	scrape component surface with metal tool

No.	Source	Interval; verdict	Label and archived note
16	003/w005/v00004	00:04:13-00:04:29; C	strike metal casting with hammer
17	003/w005/v00004	00:15:47-00:15:53; C	scoop loose parts from bulk bin
18	003/w010/v00000	00:11:45-00:12:05; C	squeeze binding paste from piping bag
19	003/w001/v00001	00:05:45-00:06:12; C	fill metal gap with dark clay
20	003/w010/v00003	00:17:12-00:17:29; C	load pre-manufactured insert into mold
21	003/w005/v00000	00:03:59-00:07:18; C	strike metal casting with hammer
22	004/w007/v00001	00:14:07-00:14:36; C	Iron folded garment with steam iron
23	004/w006/v00004	00:16:20-00:16:29; C	Unfold sewn piece to inspect seam tension
24	004/w007/v00001	00:07:22-00:07:24; C	Snap large cardboard sheet in half
25	004/w006/v00002	00:17:00-00:17:03; C	Snip loose threads with small clippers
26	004/w004/v00005	00:14:08-00:14:16; C	Push finished fabric chain to table edge
27	004/w003/v00000	00:00:00-00:01:00; C	Hem large fabric piece in short bursts; merges 5 s stand-up interruption
28	004/w005/v00001	00:08:44-00:15:18; C	Sew short straight seam; aggregated repetition
29	005/w004/v00002	00:04:43-00:04:51; C	brush metal shavings off lathe tool post
30	005/w001/v00000	00:06:26-00:06:36; C	position metallic nameplate on motor housing; split of ten-motor loop entry
31	005/w003/v00000	00:08:20-00:08:27; C	advance cutting tool via carriage handwheels
32	005/w008/v00001	00:19:10-00:19:18; P	arrange parts manually inside deep wire mesh bin; tail of composite transport entry
33	005/w006/v00000	00:06:03-00:06:10; C	initiate milling machine cycle via control panel
34	005/w003/v00000	00:07:22-00:07:29; C	advance cutting tool via carriage handwheels
35	005/w004/v00000	00:02:23-00:02:33; C	apply adhesive label to motor housing
36	006/w003/v00002	00:12:28-00:12:38; C	lift empty mold box onto pattern plate; "onto the machine"
37	006/w001/v00004	00:05:08-00:06:01; C	sift through mixed small parts on floor; floor bin
38	006/w001/v00002	00:08:42-00:08:58; C	guide upper mold half onto lower mold half
39	006/w003/v00003	00:11:39-00:11:55; C	pack casting sand into mold frame
40	006/w001/v00002	00:11:00-00:11:07; P	guide upper mold half onto lower mold half; transcript vague: "helps with a mold"
41	006/w003/v00003	00:04:24-00:04:36; C	level powder surface with wooden scraper
42	006/w003/v00002	00:16:15-00:16:30; C	pack casting sand into mold frame

Appendix E.2. Archived transcript-entry judgments

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No.	Source	Interval; verdict	Label and archived note
1	001/w008/v00010	00:12:00-00:12:51; COV	continues repetitive assembly
2	001/w004/v00007	00:05:22-00:05:39; COV	tears open bag of new metal parts
3	001/w002/v00003	00:01:07-00:01:12; COV	staples bag shut, sets aside
4	001/w010/v00002	00:11:22-00:12:05; COV	aligns hinged-clip rail on plate in jig
5	001/w007/v00004	00:15:43-00:16:41; COV	resumes stamping operation
6	001/w008/v00002	00:08:46-00:08:49; COV	camera tilt (overlapped by longer event)
7	001/w003/v00005	00:09:10-00:09:15; COV	places assembled parts onto cardboard
8	002/w005/v00009	00:20:10-00:20:13; COV	places cardboard piece on t-shirt
9	002/w003/v00000	00:09:16-00:09:44; COV	fetches and aligns garment stacks
10	002/w010/v00001	11:51:00-12:59:00; MISS	aligning stack of pink fabric pieces (archive interprets this as truncation; that cause is unverified)
11	002/w005/v00003	00:03:38-00:03:47; COV	irons folded fabric
12	002/w005/v00009	00:09:44-00:09:49; COV	irons t-shirt back
13	002/w004/v00006	00:13:17-00:13:20; EXC	short walking sequence
14	002/w002/v00009	00:11:01-00:12:35; COV	two workers fill sack with garments
15	003/w001/v00002	00:18:12-00:18:42; COV	continues scraping task
16	003/w009/v00003	00:19:35-00:19:42; COV	empties box onto tray
17	003/w004/v00002	00:02:58-00:03:08; COV	applies white paste with brush
18	003/w004/v00007	00:19:45-00:19:55; COV	continues filing metal part
19	003/w002/v00000	00:18:30-00:18:50; MISS	fetches empty tray, stages tied bundles
20	003/w004/v00002	00:06:27-00:06:33; COV	brushes paste onto part
21	003/w002/v00004	00:02:48-00:02:51; COV	drops debris into box
22	004/w006/v00004	00:17:28-00:17:34; COV	checks seam, folds fabric
23	004/w005/v00003	00:00:40-00:00:48; COV	prepares fabric edge for sewing
24	004/w009/v00001	00:09:44-00:09:46; COV	drops packaged stack in box
25	004/w006/v00004	00:08:12-00:08:29; COV	aligns fabric, resumes sewing
26	004/w003/v00000	00:19:50-00:20:00; COV	resumes continuous sewing
27	004/w006/v00004	00:11:38-00:11:43; EXC	checks phone briefly
28	004/w006/v00004	00:01:24-00:01:40; COV	sews another section
29	005/w002/v00000	00:08:04-00:08:33; EXC	brief walk around workbench, returns
30	005/w008/v00000	00:01:35-00:01:42; EXC	camera shake while walking
31	005/w003/v00000	00:05:32-00:05:40; COV	tightens fixture with wrench
32	005/w008/v00000	00:11:36-00:11:45; MISS	standing, inspecting part (borderline operational)

No.	Source	Interval; verdict	Label and archived note
33	005/w011/v00000	00:08:56-00:09:22; MISS	presses control-panel buttons, consults sheet
34	005/w008/v00000	00:05:32-00:05:54; COV	lifts bin with pallet jack, transports
35	005/w006/v00001	00:02:00-00:04:00; COV	workers stack red components into wire basket
36	006/w001/v00002	00:08:58-00:09:12; COV	waits with spray gun, sprays mold
37	006/w007/v00000	00:03:13-00:03:22; COV	hammers casting apart
38	006/w007/v00000	00:15:10-00:15:23; COV	hammers casting, scrapes sand
39	006/w001/v00007	00:11:57-00:12:03; COV	blows air onto pattern plate
40	006/w005/v00001	00:02:47-00:02:55; COV	carries mold, operates control panel
41	006/w001/v00008	00:19:19-00:19:32; COV	flips pattern plate, air-cleans
42	006/w001/v00007	00:09:24-00:09:35; COV	two workers move sand mold with lifting clamp

Appendix F. Full Extraction Prompts

These are the four released prompt templates, with punctuation normalized for display. The source files and their hashes remain in the resource. They specify requested model behavior, not guaranteed output properties. Placeholder braces are filled by the scripts; additional context is constructed as described in Table 4.

Appendix F.1. S1: Video transcription

```
Please transcribe the attached egocentric video into a very detailed chronological text description.

I do not want a formal event log, structured schema, activity labels, or any kind of post-processing. I
only want plain text entries, each beginning with a timestamp range, followed by a detailed description
of what happens in that time span.

Requirements:
- Format every entry as:
  [HH:MM:SS - HH:MM:SS] description
- Follow the video strictly from start to end.
- Use short timestamp intervals whenever something changes.
- Be highly detailed, but do not repeat the same type of behavior again and again when nothing changes.
- If the same action continues in the same way, with the same context and no relevant difference, merge
it into one longer timestamped entry instead of splitting it into many repetitive entries.
- Only create a new entry when something meaningful changes, such as:
  - the action changes
  - a new object is handled
  - the person changes location
  - attention shifts to something else
  - another person becomes involved
  - something unusual, problematic, interrupted, or different happens
- Capture micro-actions when visible, such as reaching, grasping, lifting, placing, opening, closing,
turning, walking, pausing, looking, adjusting, and interacting with objects or people.
- Describe what is visible in the egocentric view, including hand movements, object handling, scene
changes, and shifts of attention.
- If a repeated behavior continues in exactly the same way, summarize that repeated stretch once with an
 appropriate timestamp range instead of listing each repetition separately.
- If the scene is partially unclear, mention the uncertainty briefly instead of guessing.
- Do not use tables, bullets, JSON, labels, categories, or analysis language.
- Do not convert the result into an event log.
- Do not add any summary at the end.
- Only provide the timestamped descriptive transcription.

The output should read like a detailed time-aligned narration of the video, not like a structured
annotation file.

Example of the intended compression:
- Good:
  [00:03:10 - 00:03:42] The wearer continues wiping the same section of the table with repeated back and
 forth hand movements. No new object or interruption appears.
- Not desired:
  [00:03:10 - 00:03:14] The wearer wipes the table.
  [00:03:14 - 00:03:18] The wearer continues wiping the table.
  [00:03:18 - 00:03:22] The wearer continues wiping the table.

Please produce the full transcription in this style for the entire video.
```

Appendix F.2. S2: Worker summary

```
You are given multiple detailed chronological transcriptions from egocentric factory videos recorded by
the same worker.

Your task is to synthesize those transcriptions into a process-focused report about what work is being
done and what process-level optimization opportunities are visible.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

Important output constraints:
- Do not mention the factory identifier, worker identifier, file names, file paths, transcript numbers,
video numbers, or any other source identifiers.
- Do not mention timestamps unless they are absolutely necessary, which they normally are not.
- Do not quote the transcripts.
- Do not write a per-video recap. Synthesize across all provided transcriptions.

Required focus:
- Identify the process or processes performed by the worker.
- If the worker appears to perform more than one distinct process, separate them clearly.
- Describe the operational flow in enough detail that someone can understand how the work is carried out.

- Include relevant details such as setup, repeated action loops, machine interaction, material handling,
 part movement, quality checks, cleaning, waiting, transport, handoffs, and interaction with nearby
stations when visible.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly
instead of inventing details.

Optimization constraints:
- Only suggest process optimizations that are visible or reasonably inferable from the observed workflow.

- Focus on process design, workstation layout, batching, staging, transport, sequencing, buffering,
machine idle time, tool/material presentation, inspection placement, handoffs, or coordination between
steps.
- Do not suggest worker-level optimizations such as "work faster", "be more careful", "improve
discipline", "train the worker better", or similar people-focused advice.
- If no credible process optimization is evident, say so.
- When an optimization is inferential rather than directly observed, label it clearly as an inference.

Write the output in plain text with exactly these section headings:

Process Summary
Detailed Process Breakdown
Possible Process Optimizations
Evidence Limits
```

Appendix F.3. S3: Factory report and vocabularies

```
You are given multiple worker-level process summaries from the same factory.

Your task has two goals:
1. Produce a factory-level report that restates the worker process summaries in a compact but
informative way, preserving which worker each summary belongs to.
2. Produce a long, detailed, process-mining-oriented activity catalogue for the factory that can help
with downstream process optimization.

Context:
- Factory identifier: {factory}

Important constraints:
- Do not mention file names, file paths, source identifiers, transcript numbers, or video numbers.
- Do not quote the source summaries.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly
instead of inventing details.
- Synthesize across workers. Do not merely copy the worker summaries back verbatim.

Worker summary requirements:
- Include every worker provided in the input.
- Keep each worker summary concise but useful.
- Preserve the worker identifier for each summary, because the output is factory-level.
```

```
- Focus on the main process(es), major material flows, handoffs, machine interactions, transport, inspection, waiting, cleaning, batching, and coordination patterns that matter operationally.

Factory process label requirements:
- Explicitly provide one raw deduplicated list of processes executed across the different workers in the factory.
- Express each process as a short label only.
- Keep process labels concise and comparable across the factory.
- Prefer labels such as "metal stamping", "box assembly", "garment bagging", "part transport", "inspection", or "machine cleaning".
- Distinguish clearly between materially different processes, but do not create unnecessary micro-variants.
- Do not group the labels by worker.
- Do not repeat the same label for multiple workers.

Activity catalogue requirements:
- The catalogue must be long and detailed.
- Each activity must be a short string only, suitable as a candidate event label for process mining.
- Prefer specific verb-object phrases such as "load raw blank into press" rather than generic labels such as "operate machine".
- Include not only value-adding operations, but also transport, staging, waiting, inspection, rework-like handling, setup, cleanup, handoff, documentation, and coordination activities when visible.
- Keep distinct activity variants separate when they matter for process analysis.
- Avoid duplicate or near-duplicate labels.
- Avoid full sentences, explanations, numbering inside the activity text, or extra commentary next to the activity.
- If an activity is not clearly supported, do not invent it.

Optimization relevance requirements:
- Favor activity labels that would help reveal bottlenecks, queues, machine idle time, material handling burden, unnecessary motion, batching behavior, rework, inspection points, coordination dependencies, and handoff structure.

Write the output in plain text with exactly these section headings:

Factory Process Overview
Worker Process Summaries
Factory Process Labels
Factory Activity Catalogue
Optimization-Relevant Process Observations
Evidence Limits

Formatting rules for the output:
- Under "Worker Process Summaries", use one line per worker in the form:
  worker_xxx: summary
- Under "Factory Process Labels", use one bullet per process label.
- Each bullet in "Factory Process Labels" must contain only a short process label and nothing else.
- Under "Factory Activity Catalogue", use one bullet per activity.
- Each bullet in "Factory Activity Catalogue" must contain only a short activity string and nothing else.
```

Appendix F.4. S4: Event extraction

```
You are converting one timestamped raw factory annotation into a structured event list for process mining.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

You will be given:
- the allowed process labels for the factory
- the allowed activity labels for the factory
- an optional worker summary for additional context
- one raw annotation transcript with timestamp ranges measured from the start of the video

Your job:
- Convert the raw annotation into a JSON object with one key, ‘events‘, whose value is a list of event objects.
- Each event must correspond to one operationally meaningful event segment.
- Use the transcript timestamps as the main basis for the event boundaries.
- If a transcript interval contains one repeated stable activity, keep it as one longer event.
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- If a transcript interval clearly contains multiple different events, you may split it into multiple estimated events.
- Keep the events in chronological order.

Strict label constraints:
- The value of ‘activity‘ must be exactly one label from the allowed factory activity list.
- The value of ‘process‘ must be exactly one label from the allowed factory process list.
- Do not invent new activity labels.
- Do not invent new process labels.
- If multiple labels are plausible, choose the closest allowed label.
- Be especially careful to respect these two controlled vocabularies exactly.

Output schema:
- Return JSON only.
- Return a top-level JSON object with exactly one key: ‘events‘.
- The value of ‘events‘ must be a JSON array.
- Each array item must be an object with exactly these keys:
  - ‘estimated_start_time‘
  - ‘estimated_end_time‘
  - ‘activity‘
  - ‘process‘
- ‘estimated_start_time‘ and ‘estimated_end_time‘ must be strings in ‘HH:MM:SS‘ format, measured from the start of the video.

Additional rules:
- Include only process-relevant operational events. Ignore purely accidental camera motion or camera removal unless it directly reflects a process-relevant activity.
- Do not output markdown fences.
- Do not output explanations, notes, comments, or any text before or after the JSON object.
- If no valid operational events can be extracted, return ‘{"events":[]}‘.
```